Week-12

1. What does a ‘guarded’ attribute mean in a language model’s representation space?
    a. The attribute can only be classified by specific linear probes.
    b. The attribute requires non-linear transformations to be classified.
    c. The attribute is heavily emphasized during the initial pre-training process.
    d. The attribute is entirely impossible to classify based on the representation.

    Explanation: An attribute Z is considered guarded if we can't classify along that attribute.

2. What is the primary objective of modifying vectors from one distribution (e.g., toxic to non-toxic) to look like another?
    a. To increase the computational cost of generating toxic content.
    b. To guard an attribute with the smallest possible change to unrelated semantics.
    c. To fundamentally alter the underlying vocabulary of the model.
    d. To completely randomize the vector outputs of specific trigger tokens.

    Explanation: We want to guard an attribute with the smallest change possible to preserve semantics unrelated to Z.

3. According to the lecture, what is the implication behind using Affine Steering?
    a. It is a no-gradient and cheap way to control model generation.
    b. It fine-tunes the model to effectively remove bias using minimal training data.
    c. It is a highly interpretable, gradient-based optimization method.
    d. It depends heavily on reinforcement learning and human feedback.

    Explanation: Affine Steering is a no-gradient and cheap way to control generation and to de-bias.

4. How does the size of a conventional cell complex scale as the size of the graph increases?
    a. It scales inversely with respect to graph size.
    b. It scales quadratically with respect to graph size.
    c. It scales logarithmically with respect to graph size.
    d. It scales linearly with respect to graph size.

    Explanation: The size of a conventional cell complex will scale quadratically as the size of the graph increases.

5. In the molecule toxicity detection task, why might a standard Graph Neural Network Explainer fail to identify the correct explanation?
    a. It removes all nodes that have a high edge degree.
    b. It processes node features too slowly for a chemical dataset.
    c. It cannot capture specific multi-node groupwise interactions.
    d. It assigns zero weight to all the chemical bonds by default.

    Explanation: A limitation of graphs is that they can only model pairwise relationships between nodes.

6. How does FORGE reduce the computational complexity of the cell complex to a linear scale?
    a. By randomly sampling only 10% of the original graph edges.
    b. By strategically removing specific types of boundary relations.
    c. By completely deleting all nodes with a degree higher than a threshold degree.
    d. By executing the explainer algorithm on highly compressed latent space.

    Explanation: By strategically removing specific types of boundary relations, we can bring this down to linear complexity.

7. How are cycles structurally represented in the cell complex?
    a. They are represented as 0-cells.
    b. They are represented as 1-cells.
    c. They are represented as 2-cells.
    d. They are represented as 3-cells.

    Explanation: Cycles are 2-cells, which are represented as augmented nodes connected to all edges that make up that cycle.

8. Which of the following are examples of graph explainers? **(Select all that apply.)**
    a. Predicting the next token in a language model
    b. Toxicity detection of molecules
    c. Affine concept erasure
    d. Drug prediction
    e. Fraud detection

    Explanation: Graph explainers are important in toxicity detection of molecules, drug prediction, fraud detection, etc.

9. Which of the following methods benefits the most from the FORGE framework?
    a. Attention-based methods
    b. Surrogate-based methods
    c. Gradient-based methods
    d. Perturbation-based methods

    Explanation: Perturbation-based methods benefit the most from FORGE.

10. Based on the mathematical definitions of cell complexes presented in the FORGE slides, which of the following statements are correct? **(Select all that apply.)**
    a. A 0-cell corresponds to a vertex, while a 1-cell corresponds to an edge in traditional graphs.
    b. The 1-skeleton $X^{(1)}$ of a cell complex consists exclusively of 1-dimensional cells.
    c. Vertical boundary relations connect two cells of the same dimension that share a common boundary.
    d. Boundary relations in the framework are restricted to be undirected, meaning $\sigma_{c_1,c_2} = \sigma_{c_2,c_1}$.
    e. A p-cell $c^{(p)}$ is considered a face or boundary of a $(p+1)$-cell $c^{(p+1)}$ if the set of points composing $c^{(p)}$ is a subset of those composing $c^{(p+1)}$.

Explanation: A - Definition 3.1

B - Definition 3.4 states that $X^{(p)}$ is a subcomplex of cells of dimension at most p, meaning both 0- and 1-cells.

C - Definition 3.5 states horizontal boundary relations as linking two p-cells of the *same* dimension, whereas vertical boundary relations link cells of *different* dimensions.

D - Definition 3.5

E - Definition 3.2